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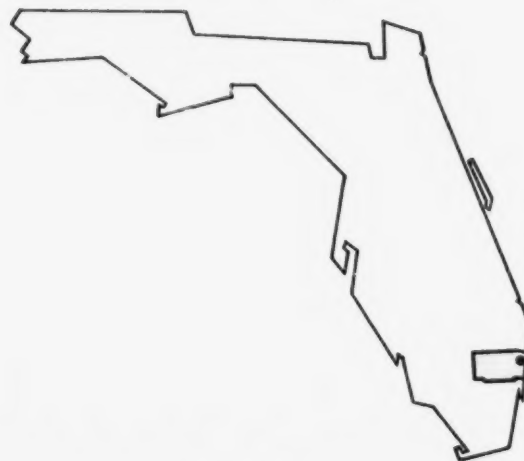
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FLOOD INSURANCE STUDY

WAVE
HEIGHT
ANALYSIS



VILLAGE OF
SEA RANCH LAKES,
FLORIDA
BROWARD COUNTY



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Federal Emergency Management Agency

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ANALYSIS OF WAVE HEIGHTS
FOR
VILLAGE OF SEA RANCH LAKES, BROWARD COUNTY, FLORIDA

FEDERAL EMERGENCY MANAGEMENT AGENCY

TABLE OF CONTENTS

	Page
1.0 <u>INTRODUCTION</u>	1
1.1 Background and Purpose.....	1
2.0 <u>INVESTIGATIONS</u>	1
2.1 Previous Studies.....	1
2.2 Data Collection and Review.....	2
2.3 Wave Height Analysis.....	3
2.4 Results.....	6
3.0 <u>FLOOD PLAIN MANAGEMENT APPLICATIONS</u>	6
3.1 Flood Boundaries.....	6
3.2 Base Flood Elevations.....	8
3.3 Velocity Zones.....	8
4.0 <u>INSURANCE APPLICATION</u>	8
4.1 Flood Hazard Factors.....	9
4.2 Flood Insurance Zones.....	9
4.3 Flood Insurance Rate Map.....	11
5.0 <u>OTHER STUDIES</u>	11
6.0 <u>REFERENCES</u>	11

FIGURES

Figure 1 - Transect Location Map.....	4
Figure 2 - Typical Transect Schematic.....	7

TABLES

Table 1 - Transect Locations, Stillwater Elevations, and Maximum Wave Crest Elevations.....	5
Table 2 - Flood Insurance Zone Data.....	10

EXHIBITS

Exhibit 1 - Flood Insurance Rate Map	
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1.0 INTRODUCTION

1.1 Background and Purpose

The Federal Emergency Management Agency recently adopted recommendations by the National Academy of Sciences to include prediction of wave heights in Flood Insurance Studies for coastal communities subject to storm surge flooding, and to report the estimated wave crest elevations as the base flood elevations on Flood Insurance Rate Maps (FIRMs).

Previously, FIRMs were produced showing only the stillwater storm-surge elevations due to the lack of a suitable and generally applicable methodology for estimating the wave crest elevations associated with storm surges. These stillwater elevations were subsequently stipulated in community flood plain management ordinances as the minimum elevation of the first floor of new construction. Communities and individuals had to consider the additional hazards of velocity waters and wave action on an ad hoc basis. Because there has been a pronounced tendency for buildings to be constructed only to meet minimum standards, without consideration of the additional hazard due to wave height, increasing numbers of people could unknowingly be accepting a high degree of flood-related personal and property risk in coastal areas subject to wave action. Therefore, the Federal Emergency Management Agency has pursued the development of a suitable methodology for estimating the wave crest elevations associated with storm surges. The recent development of such a methodology by the National Academy of Sciences (Reference 1) has led to the adoption of wave crest elevations for use as the base flood elevations in coastal communities.

Among the most severe storms to pass through Broward County were those of September 1926, September 1928, September 1947, and August 1964. The hurricane of September 1926, was one of the four most destructive hurricanes of the 20th century. The Hollywood-Fort Lauderdale area suffered extensive damage due to tidal flooding and wave action. The hurricane of September 1947 resulted in widespread flooding and extensive crop damage. Winds of up to 150 miles per hour were reported at Pompano Beach.

The Village of Sea Ranch Lakes, Broward County, Florida, is located on the open coast and is subject to flooding from tidal surges associated with hurricanes. The purpose of this study is to revise the FIRM for Sea Ranch Lakes to include the effects of wave action associated with flooding from the Atlantic Ocean at the open coast and at the Intracoastal Waterway.

2.0 INVESTIGATIONS

2.1 Previous Studies

Stillwater elevations for hurricane surges with 10-, 50-, 100-, and 500-year return periods were determined by the U.S. Army Corps of Engineers, Jacksonville District, for the open coast of Sea Ranch Lakes. The 100-year stillwater

elevations were used as base flood elevations for the Flood Insurance Study and the Flood Insurance Rate Map that has been prepared for Sea Ranch Lakes (References 2 and 3).

The determination of the open coast surge was based on the Bathystropic Storm Tide Technique utilizing a computer program developed by the U.S. Army Corps of Engineers Coastal Engineering Research Center (Reference 4). This technique involves selection of wind velocities and direction, wind stress coefficients, bottom friction factors, radius of maximum winds, central pressure index, and speed of movement of the storm.

In computing the 10-, 50-, 100-, and 500-year surges in the study area, hypothetical hurricanes encountered along the coast near the study area were defined using various National Weather Service publications (References 5 and 6) to select parameters for use as input into the computer program. The computer program is a dynamic model which utilizes the effects of wind on water by principles of fluid and wave mechanics. The program calculates the open coast surge elevation in hourly increments and develops a storm hydrograph which includes the peak elevation of the storm along the open coast.

Determination of the flood elevations along the Intracoastal Waterway in the study area was based on a flood routing technique. This technique involved using the open coast surge values at Hillsboro Inlet and Port Everglades and hydraulically routing those surges along the Intracoastal Waterway. Cross sections for this analysis were determined from detailed soundings of the Intracoastal Waterway from Hillsboro Inlet to Port Everglades by the U.S. Army Corps of Engineers, and from U.S. Geological Survey topographic maps.

Stillwater elevations used for this wave height analysis were accepted from the above mentioned studies. This study is authoritative for purposes of the National Flood Insurance Program. Data presented herein supersede all previous determinations.

2.2 Data Collection and Review

All available source data applicable for the wave height analysis were collected and reviewed. Because wave height calculations are based on such parameters as the size and density of vegetation, natural barriers (such as sand dunes), buildings, and other manmade structures, it was necessary to obtain detailed information on the physical and cultural features of the study area.

Officials of the Village of Sea Ranch Lakes and Broward County were contacted for maps and other possible data sources to be used in the analysis. In addition, the Florida Department of Natural Resources, Bureau of Beaches and Shores; the Florida Department of Transportation; the U.S. Army Corps of Engineers, Jacksonville District; the U.S. Fish and Wildlife Service; Arthur V. Strock and Associates, Inc.; and the National Ocean Survey were contacted, and all applicable data sources were obtained.

The principal source materials used for the wave height analysis are described below.

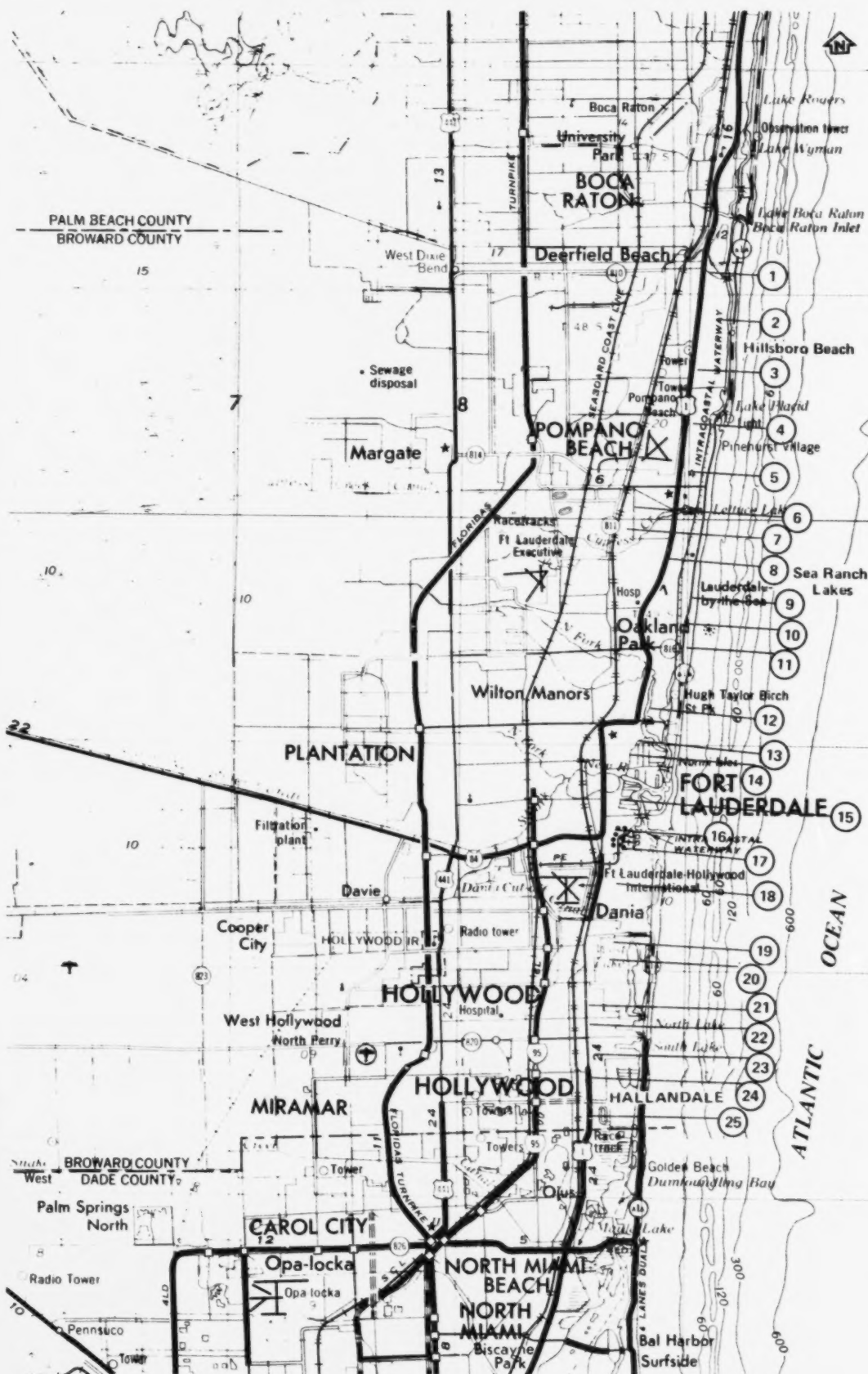
- a. Topographic data (ground elevations, seawall elevations, and locations of physical features) were determined using U.S. Geological Survey topographic maps at a scale of 1:24,000, with a contour interval of 5 feet (Reference 7); National Ocean Survey storm evacuation maps at a scale of 1:62,500, with a contour interval of 5 feet (Reference 8); beach profiles of the immediate Atlantic coast area obtained from the Florida Department of Natural Resources (Reference 9) and the U.S. Army Corps of Engineers (Reference 10), and Arthur V. Strock and Associates, Inc., Coastal Structures Inventory maps (Reference 11).
- b. Recent stereoscopic aerial photography at a scale of 1:12,000 was used to determine vegetation parameters and to supplement the topographic maps for locations of physical features (Reference 12).

2.3 Wave Height Analysis

The methodology for analyzing the effects of wave heights associated with coastal storm surge flooding is described in the National Academy of Sciences report (Reference 1). This method is based on the following major concepts. First, depth-limited waves in shallow water reach a maximum breaking height that is equal to 0.78 times the stillwater depth. The wave crest is 70 percent of the total wave height above the stillwater level. The second major concept is that wave height may be diminished by dissipation of energy due to the presence of obstructions, such as sand dunes, dikes and seawalls, buildings, and vegetation. The amount of energy dissipation is a function of the physical characteristics of the obstruction and is determined by procedures prescribed in Reference 1. The third major concept is that wave height can be regenerated in open fetch areas due to the transfer of wind energy to the water. This added energy is related to fetch length and depth.

Wave heights were computed along transects (cross section lines) in accordance with the Users Manual for Wave Height Analysis (Reference 13). The transects were located along the coastline as shown in Figure 1. They were located with consideration given to the physical and cultural characteristics of the land so that they would closely represent conditions in their locality. Transects were spaced close together in areas of complex topography and dense development. In areas having more uniform characteristics, they were spaced at larger intervals. It was also necessary to locate transects in areas where unique flooding existed and in areas where computed wave heights varied significantly between adjacent transects.

Each transect was taken perpendicular to the shoreline and extended to the inland limit of tidal flooding. Along each transect, wave heights and elevations were computed considering the combined effects of changes in ground elevation, vegetation, and physical features. The stillwater elevations for the 100-year flood were used as the starting elevations for these computations. Wave heights were calculated to the nearest 0.1 foot, and wave elevations were determined at whole-foot increments along the transects. The location of the 3-foot breaking wave for determining the terminus of the V Zone (area with velocity wave action) was also computed at each transect. Table 1 provides a listing of the transect locations and stillwater elevations, as well as maximum wave crest elevations in the vicinity of Sea Ranch Lakes.



TRANSECT LOCATION MAP

FEDERAL EMERGENCY MANAGEMENT AGENCY

BROWARD COUNTY, FL

FIGURE 1

Table 1.
Transect Locations, Stillwater Elevations, and Maximum Wave Crest Elevations

Transect	Location	Elevation (Feet NGVD)	
		Stillwater	Wave Crest
7	Broward County - From the Atlantic coastline westward across the Intracoastal Waterway flood plain approximately 500 feet south of the intersection of South Ocean Boulevard and Sunset Lane	8.0 ¹	12 ¹
8	Broward County/Sea Ranch Lakes - From the Atlantic coastline westward across the Intracoastal Waterway flood plain approximately 500 feet north of the intersection of South Ocean Boulevard (AlA) and Gate House Road in Sea Ranch Lakes	6.0 ²	6 ²
¹ Atlantic Ocean, open coast			
² Intracoastal Waterway			

Figure 2 is a profile for a typical transect illustrating the effects of energy dissipation and regeneration on a wave as it moves inland. This figure shows the wave elevations being decreased by obstructions, such as buildings, vegetation, and rising ground elevations, and being increased by open, unobstructed wind fetches. Actual wave conditions in Sea Ranch Lakes may not necessarily include all the situations illustrated in Figure 2.

After analyzing wave heights along each transect, wave elevations were interpolated between transects. Various source data were used in the interpolation, including topographic maps (References 6 and 7), beach profiles (Reference 9), and aerial photographs (Reference 12), along with engineering judgment. Controlling features affecting the elevations were identified and considered in relation to their positions at a particular transect and their variation between transects.

2.4 Results

The only area affected by wave action in Sea Ranch Lakes is along the open coast. Wave heights greater than 3 feet are diminished at the beach face by rising ground elevations. The community is provided additional protection by a seawall extending the entire length of the coast of Sea Ranch Lakes. This seawall is above the 100-year tidal flood plain and, therefore, was not considered to effect wave heights in the community. The maximum wave crest elevation at the Atlantic Ocean shoreline is 12 feet.

The narrowness of the Intracoastal Waterway in conjunction with the dense development of the area prevents any wave generation across the Intracoastal Waterway flood plain.

3.0 FLOOD PLAIN MANAGEMENT APPLICATIONS

A prime purpose of the National Flood Insurance Program is to encourage local governments to adopt sound flood plain management programs designed to reduce future flood losses. The FIRM for Sea Ranch Lakes has been revised to incorporate the latest available information, including wave height data, to assist this community in developing the most appropriate and effective flood plain management measures.

3.1 Flood Boundaries

In order to provide a national standard without regional discrimination, the 100-year flood has been adopted by the Federal Emergency Management Agency as the base flood for purposes of flood plain management. This flood has a 1 percent chance of being equalled or exceeded each year and is expected to be exceeded once on the average during any 100-year period. The risk of having a flood of this magnitude or greater increases when periods longer than 1 year are considered. For example, over a 30-year period, there is a 26 percent chance of experiencing a flood equal to or greater than the 100-year flood. The 500-year flood plain is also shown on the FIRM to indicate areas of moderate flood hazards.

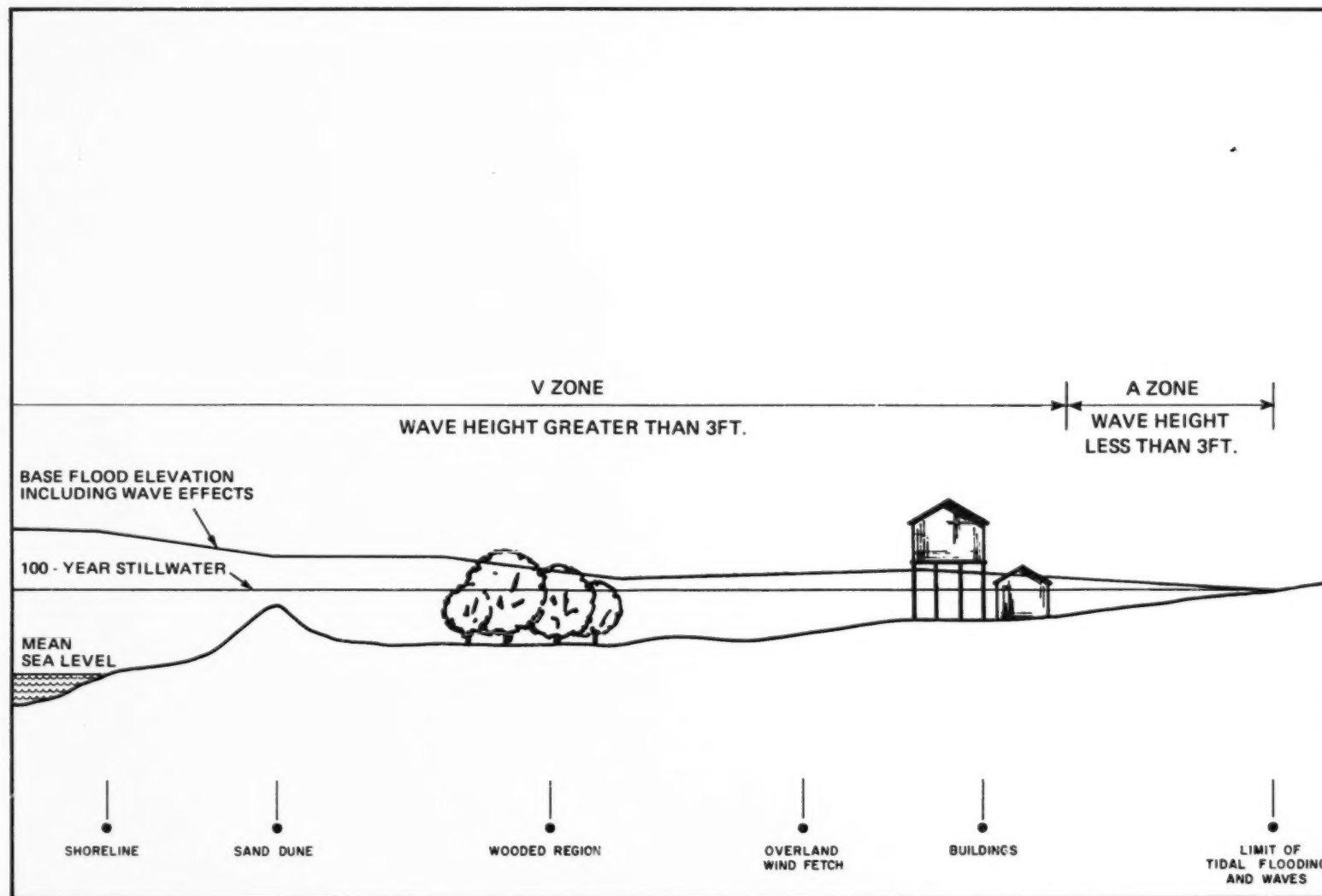


FIGURE 2
TYPICAL TRANSECT SCHEMATIC

Areas inundated by the 100-year flood are shown as A and V Zones on the community's FIRM. It is in these areas that the Federal Emergency Management Agency requires local communities to exercise flood plain management measures as a condition for participation in the National Flood Insurance Program.

3.2 Base Flood Elevations

Areas within the communities studied by detailed engineering methods have base flood elevations established in A and V Zones. These are the elevations of the base (100-year) flood relative to the National Geodetic Vertical Datum of 1929. In coastal areas affected by wave action, base flood elevations are generally maximum at the open coast shoreline. These elevations generally decrease in a landward direction at a rate dependent on the presence of obstructions capable of dissipating the wave energy. Where possible, changes in base flood elevations have been shown in 1-foot increments on the FIRMs. However, where the scale did not permit, 2- or 3-foot increments were sometimes used. Base flood elevations shown in the wave action areas represent the average elevation within the zone. Current program regulations generally require that all new construction be elevated such that the first floor, including basement, is above the base flood elevation in A and V Zones.

3.3 Velocity Zones

The U.S. Army Corps of Engineers (Reference 14) has established the 3-foot breaking wave as the criterion for identifying coastal high hazard zones. This was based on a study of wave action effects on structures. This criterion has been adopted by the Federal Emergency Management Agency for the determination of V Zones. Because of the additional hazards associated with high-energy waves, the National Flood Insurance Program regulations require much more stringent flood plain management measures in these areas, such as elevating structures on piles or piers. In addition, insurance rates in V Zones are higher than those in A Zones with similar numerical designations.

The location of the V Zone is determined by the 3-foot breaking wave as discussed previously. The detailed analysis of wave heights performed in this study allowed a much more accurate location of the V Zone to be established. The V Zone generally extends inland to the point where the 100-year flood depth is insufficient to support a 3-foot breaking wave.

4.0 INSURANCE APPLICATION

The assignment of proper actuarial insurance rates requires that frequency and depth of flooding be estimated as accurately as possible. Because waves can add considerably to expected flood depths, it is important that insurance rates consider this additional hazard. The Federal Emergency Management Agency has developed a process to transform the data from this study into flood insurance criteria. This process includes the determination of Flood Hazard Factors and the designation of flood insurance zones.

4.1 Flood Hazard Factors

The Flood Hazard Factor (FHF) is the device used to correlate flood information with insurance rate tables. Correlations between property damage from floods and their FHF are used to set actuarial insurance premium rate tables.

The FHF is shown as a three-digit code that expresses the difference between the 10- and 100-year flood elevations to the nearest 0.5 foot. For example, if the difference between water-surface elevations of the 10- and 100-year floods is 0.7 foot, the FHF is 005; if the difference is 1.4 feet, the FHF is 015; if the difference is 5.0 feet, the FHF is 050. When the difference between the 10- and 100-year water-surface elevations is greater than 10.0 feet, the FHF is computed to the nearest foot.

4.2 Flood Insurance Zones

After wave elevations for the 100-year storm surge were determined and mapped, the study areas were divided into zones, each having a specific flood potential and FHF. Each zone was assigned one of the following flood insurance zone designations:

Zone V12:	Special Flood Hazard Areas inundated by the 100-year flood, and that have additional hazards due to velocity (wave action); base flood elevations shown, and zones subdivided according to FHF's.
Zones A5 and A10:	Special Flood Hazard Areas inundated by the 100-year flood, with base flood elevations determined and zone designations assigned according to the FHF's.
Zone B:	Areas between the Special Flood Hazard Areas and the limits of the 500-year flood plain, including areas that are protected from the 100- and 500-year floods by dike, levee, or other water control structure; and areas subject to certain types of 100-year shallow flooding where depths are less than 1.0 foot, or where the contributing drainage area is less than 1 square mile. Zone B is not subdivided.
Zone C:	Areas not subject to flooding by the 500-year flood. Zone C is not subdivided.

Table 2, "Flood Insurance Zone Data," summarizes the FHF's, flood insurance zones, and base flood elevations for each flooding source in the study area.

Table 2. Flood Insurance Zone Data

<u>Flooding Source</u>	<u>Stillwater Elevation</u>				<u>FHF</u>	<u>Zone</u>	<u>Base Flood Elevation (Feet NGVD) *</u>
	<u>10-Year</u>	<u>50-Year</u>	<u>100-Year</u>	<u>500-Year</u>			
Atlantic Ocean							
Open Coast	4.1	6.8	8.1	10.8	060	V12	10 to 12
					050	A10	8 to 10
Intracoastal Waterway	3.4	5.1	6.0	7.1	025	A5	6

*Due to map scale limitations, base flood elevations shown on the FIRM may represent average elevations for the zones depicted.

4.3 Flood Insurance Rate Map

After flood insurance zones were established for the study area, the FIRM for Sea Ranch Lakes was revised to incorporate the new zone information. This map (Exhibit 1) contains the official delineation of flood insurance zones and base flood elevations.

In preparing the revised FIRM, a current community map (Reference 15) was used to update the corporate limits and some cultural features.

5.0 OTHER STUDIES

FIRMS for the adjacent communities of Fort Lauderdale, Lauderdale-by-the-Sea, and Broward County, are being revised concurrently with Sea Ranch Lakes to include wave height analysis. The revised studies for these communities will be in agreement with this study, except for differences in zone numbers between Sea Ranch Lakes and the adjacent communities. These differences are attributed to minor variations in surge elevations used in the analyses.

6.0 REFERENCES

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